Decision Tree using Gain Ratio (C4.5)

Unit -3

Why Gain Ratio?

- The information gain measure is biased toward tests with many outcomes.
- That is, it prefers to select attributes having a large number of values.
- For example, consider an attribute that acts as a unique identifier such as product ID.
- A split on product ID would result in a large number of partitions (as many as there are values), each one containing just one tuple.
- Because each partition is pure, the information required to classify data set D based on this partitioning would be Infoproduct ID(D) = 0.
- Therefore, the information gained by partitioning on this attribute is maximal. Clearly, such a partitioning is useless for classification.
- C4.5, a successor of ID3, uses an extension to information gain known as gain ratio, which attempts to overcome this bias. It applies a kind of normalization to information gain using a “split information” .

$$\boxed{SplitInfo_A(D) = -\sum_{j=1}^{v} \frac{|D_j|}{|D|} \times \log_2\left(\frac{|D_j|}{|D|}\right).}$$

$$\boxed{GainRatio(A) = \frac{Gain(A)}{SplitInfo_A(D)}}$$

CGPA	Inter active	Practical Knowledge	Comm Skills	Job Offer
>=9	Yes	Very good	Good	Yes
>=8	No	Good	Moderate	Yes
>=9	No	Average	Poor	No
<8	No	Average	Good	No
>=8	Yes	Good	Moderate	Yes
>=9	Yes	Good	Moderate	Yes
<8	Yes	Good	Poor	No
>=9	No	Very good	Good	Yes
>=8	Yes	Good	Good	Yes
>=8	Yes	Average	Good	Yes

1. Compute class entropy

$$\text{Entropy\_Info}(T) = -\sum_{i=1}^{m} P_i \log_2 P_i$$

- Entropy(Job Offer) =

$$-\left\{ \left(7/10 \log_2 (7/10)\right) + \left(3/10 \log_2 (3/10)\right) \right\}$$

Entropy(Job Offer) = 0.8807

$$\text{Entropy\_Info}(T, A) = \sum_{i=1}^{v} \frac{|A_i|}{|T|} \times \text{Entropy\_Info}(A_i)$$

2. Calculate Entropy_Info

CGPA	Inter active	Practical Knowledge	Comm Skills	Job Offer
>=9	Yes	Very good	Good	Yes
>=8	No	Good	Moderate	Yes
>=9	No	Average	Poor	No
<8	No	Average	Good	No
>=8	Yes	Good	Moderate	Yes
>=9	Yes	Good	Moderate	Yes
<8	Yes	Good	Poor	No
>=9	No	Very good	Good	Yes
>=8	Yes	Good	Good	Yes
>=8	Yes	Average	Good	Yes

- Consider attribute CGPA.
- Entropy_Info(CGPA) =

$$\{ (4/10) ([-3/4 \log_2(3/4)] + [-1/4 \log_2(1/4)]) \} +$$

$$\{ (4/10) ([-4/4 \log_2(4/4)] + [-0/4 \log_2(0/4)]) \} +$$

$$\{ (2/10) ([-0/2 \log_2(0/2)] + [-2/2 \log_2(2/2)]) \}$$

- **Entropy_Info(CGPA) = 4/10 (0.3111+0.4997) + 0 + 0**
- **Entropy_Info(CGPA) = 0.3243**

3. Gain(Info_Gain)

$$\text{Information\_Gain}(A) = \text{Entropy\_Info}(T) - \text{Entropy\_Info}(T, A)$$

- Gain = 0.8807 – 0.3243 = 0.5564

4. Split _Info

$$SplitInfo_A(D) = -\sum_{j=1}^{v} \frac{|D_j|}{|D|} \times \log_2\left(\frac{|D_j|}{|D|}\right).$$

Split _Info = { [-(4/10) * $\log_2$(4/10)] + [-(4/10) * $\log_2$(4/10)] + [-(2/10) * $\log_2$(2/10)] }

- Split _Info = 0.5285 + 0.5285 + 0.4641

- Split _Info = 1.5211

$$GainRatio(A) = \frac{Gain(A)}{SplitInfo_A(D)}$$

5. Gain Ratio = 0.5564 / 1.5211

    Gain Ratio = 0.3658

Now calculate for rest of the attributes….

- Gain Ratio (Interactive)

- Gain Ratio (Practical Knowledge)

- Gain Ratio (Comm Skills)

- Gain Ratio (Interactive) = 0.0911 / 0.9704 = 0.0939

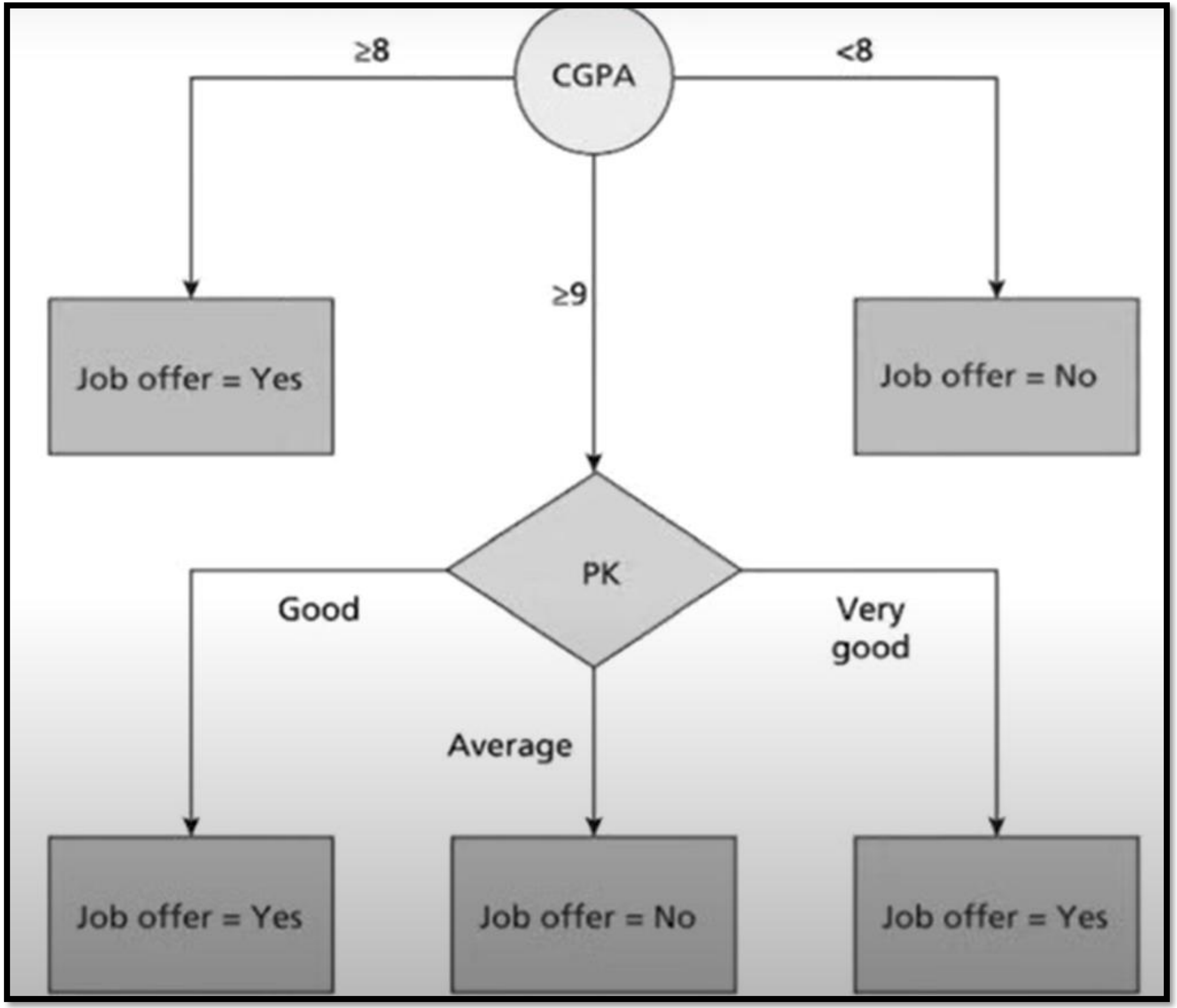
- Gain Ratio (Practical Knowledge) = 0.2448 / 1.4853
- Gain Ratio (Practical Knowledge) = 0.1648

- Gain Ratio (Comm Skills) = 0.5202 / 1.4853
- Gain Ratio (Practical Knowledge) = 0.3502

Inter active	Practical Knowledge	Comm Skills	Job Offer
Yes	Very good	Good	Yes
No	Average	Poor	No
Yes	Good	Moderate	Yes
No	Very good	Good	Yes

- Gain Ratio

- Interactive = 0.3112
- Practical Knowledge = 0.5408
- Comm Skills = 0.5408


CGPA
≥8
<8
≥9
Job offer = Yes
Job offer = No
PK
Good
Very good
Average
Job offer = Yes
Job offer = No
Job offer = Yes

PART B

A) Draw Decision Tree using ID3 algorithm for the following dataset. **(8)**

Hair	**Height**	**Weight**	**Class**
Blonde	Average	Light	No
Blonde	Tall	Average	Yes
Brown	Short	Average	Yes
Blonde	Short	Average	No
Red	Average	Heavy	No
Brown	Tall	Heavy	No
Brown	Average	Heavy	No
Blonde	Short	Light	Yes